



The Routing Table

(Illustrated from a Cisco Router)

Introduction

- Since a router connects networks by determining the best path and forwarding packets, it must know how reach its locally attached networks and remote networks
 - This information is held in the **routing table**
- A **routing table** is a type of data file that acts as a map that is used provide optimal routes between network nodes for the efficient forwarding of traffic between them and the traffic's destination
 - It is often installed on a router, networked computer or other hardware
- Routers build and maintain routing tables which should reflect current information

Routing Table Entries(Cisco Router)



Strathmore
UNIVERSITY

Command to view the routing table

```
Router#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

172.16.0.0/24 is subnetted, 3 subnets
O       172.16.1.0 [110/65] via 172.16.4.251, 00:00:44, Serial2/0
C       172.16.3.0 is directly connected, FastEthernet0/0
C       172.16.4.0 is directly connected, Serial2/0
Router#
```

**Source of
routing
information**

**Destination network
(remote or local)**

**IP address of the
interface on the next
router through which
the remote network
can be accessed**

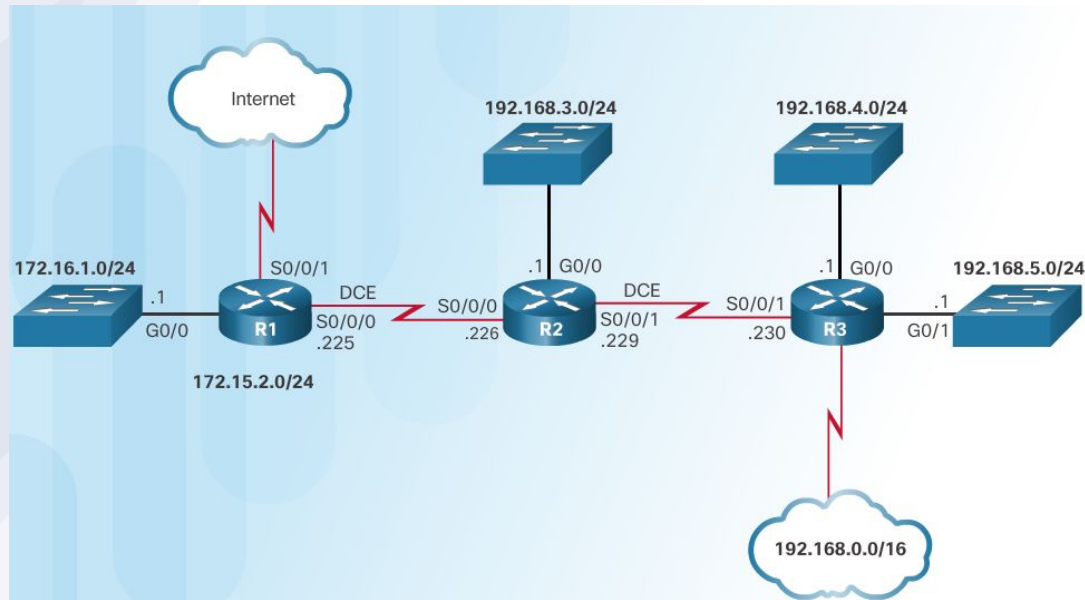
**Local interface on which
the local or remote
network can be reached on**

Parts of an IPv4 Route Entry

Routing Table Entries



Strathmore
UNIVERSITY



```
R1# show ip route | begin Gateway
Gateway of last resort is 209.165.200.234 to network 0.0.0.0

S* 0.0.0.0/0 [1/0] via 209.165.200.234, Serial0/0/1
    is directly connected, Serial0/0/1
C 172.16.0.0/16 is variably subnetted, 5 subnets, 3 masks
L 172.16.1.0/24 is directly connected, GigabitEthernet0/0
L 172.16.1.1/32 is directly connected, GigabitEthernet0/0
R 172.16.2.0/24 [120/1] via 209.165.200.226, 00:00:12, Serial0/0/0
R 172.16.3.0/24 [120/2] via 209.165.200.226, 00:00:12, Serial0/0/0
R 172.16.4.0/28 [120/2] via 209.165.200.226, 00:00:12, Serial0/0/0
R 192.168.0.0/16 [120/2] via 209.165.200.226, 00:00:03, Serial0/0/0
209.165.200.0/24 is variably subnetted, 5 subnets, 2 masks
C 209.165.200.224/30 is directly connected, Serial0/0/0
L 209.165.200.225/32 is directly connected, Serial0/0/0
R 209.165.200.228/30 [120/1] via 209.165.200.226, 00:00:12, Serial0/0/0
C 209.165.200.232/30 is directly connected, Serial0/0/1
L 209.165.200.233/30 is directly connected, Serial0/0/1
R1#
```

Routing Table for R1

Parts of an IPv4 Route Entry

Directly Connected Entries



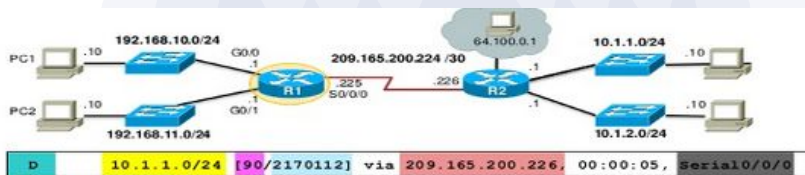
Strathmore
UNIVERSITY

- Directly Connected Networks (C) are automatically added to the routing table when the interface is configured and activated.
- Entries contain the following information:
 - Route source - how the route was learned.
 - Destination network – remote network.
 - Outgoing Interface – exit interface used to forward packets to destination.
- Other route source entries include:
 - S –Static Route
 - D – EIGRP routing protocol
 - O – OSPF routing protocol
 - R - RIP routing protocol

Route Source	Destination Network	Outgoing Interface
C	172.16.1.0/24 is directly connected,	GigabitEthernet0/0
L	172.16.1.1/32 is directly connected,	GigabitEthernet0/0

Legend

- Identifies how the network was learned by the router.
- Identifies the destination network and how it is connected.
- Identifies the interface on the router connected to the destination network.

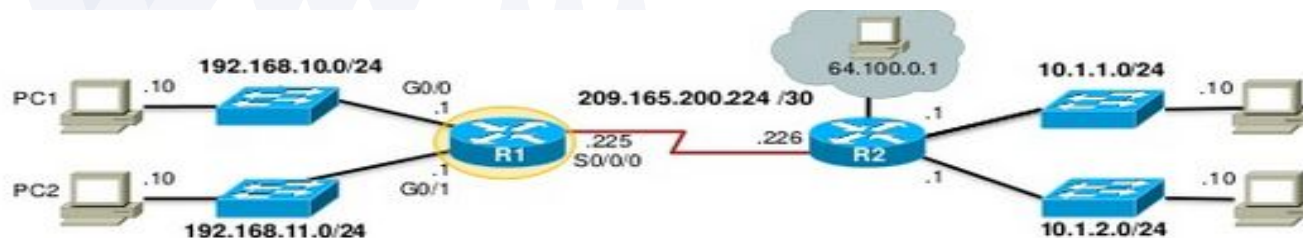


A	Identifies how the network was learned by the router.
B	Identifies the destination network.
C	Identifies the administrative distance (trustworthiness) of the route source.
D	Identifies the metric to reach the remote network.
E	Identifies the next hop IP address to reach the remote network.
F	Identifies the amount of elapsed time since the network was discovered.
G	Identifies the outgoing interface on the router to reach the destination network.

Remote Routing Table Entries (Cisco Router)



- Whenever a static route or dynamic routing protocol is configured an entry for remote network is added to the routing table.
- The entries contains the fields shown above.
- **Note:** The source of the routing will have a specific code. E.g.
 - S –Static Route
 - D – EIGRP routing protocol
 - O – OSPF routing protocol
 - R - RIP routing protocol



D	10.1.1.0/24	[90/2170112]	via	209.165.200.226,	00:00:05,	Serial0/0/0
---	-------------	--------------	-----	------------------	-----------	-------------

A	Identifies how the network was learned by the router.
B	Identifies the destination network.
C	Identifies the administrative distance (trustworthiness) of the route source.
D	Identifies the metric to reach the remote network.
E	Identifies the next hop IP address to reach the remote network.
F	Identifies the amount of elapsed time since the network was discovered.
G	Identifies the outgoing interface on the router to reach the destination network.

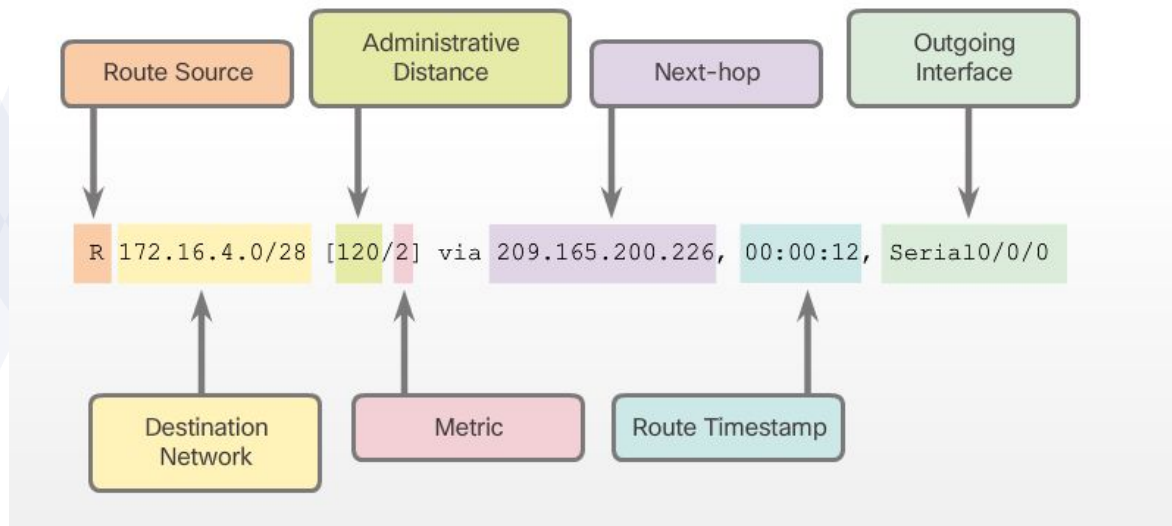
Parts of an IPv4 Route Entry

Remote Network Entries



Strathmore
UNIVERSITY

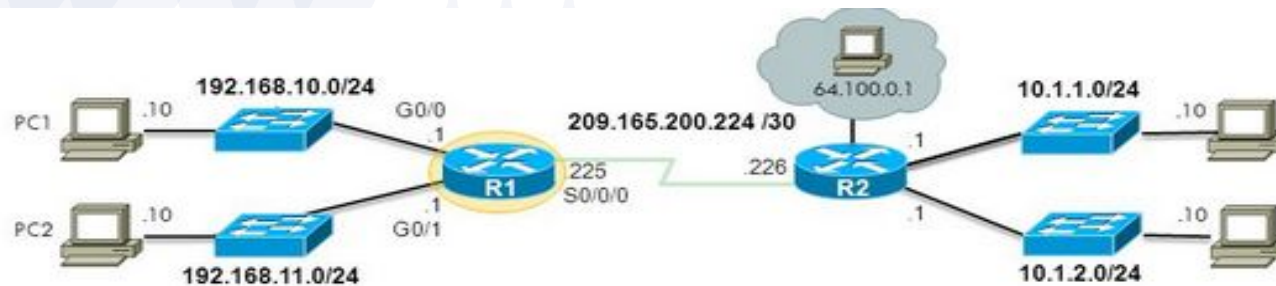
- Routes to remote networks contain the following information:
 - Route source – how route was learned
 - Destination network
 - Administrative distance (AD) - trustworthiness of the route.
 - Metric – value assigned to reach the remote network. Lower is better.
 - Next hop – IPv4 address of the next router that the packet should be forwarded to.
 - Route timestamp – time since the route was updated.
 - Outgoing interface - the exit interface to use to forward the packet



Directly Connected Entries (Cisco Router)



- Directly Connected Networks (C) are automatically added to the routing table when the interface is configured and activated
- Local (L) represents the actual IP configured on the interface.
 - This is a feature of some routers; it depends on the router's operating system
 - L always appears whenever IPv6 addresses are configured



A		B		C	
C		192.168.10.0/24 is directly connected,	GigabitEthernet0/0		
L		192.168.10.1/32 is directly connected,	GigabitEthernet0/0		

A	Identifies how the network was learned by the router.
B	Identifies the destination network and how it is connected.
C	Identifies the interface on the router connected to the destination network.

Dynamically Learned IPv4 Routes

Routing Table Terms



Strathmore
UNIVERSITY

- The routing table is a hierarchical structure that is used to speed up the lookup process when locating routes and forwarding packets.
- The hierarchy includes:
 - Ultimate Routes
 - Level 1 routes
 - Level 1 parent routes
 - Level 2 child routes

```
R1# show ip route | begin Gateway
Gateway of last resort is 209.165.200.234 to network 0.0.0.0

S*    0.0.0.0/0 [1/0] via 209.165.200.234, Serial0/0/1
        is directly connected, Serial0/0/1
        172.16.0.0/16 is variably subnetted, 5 subnets, 3 masks
C      172.16.1.0/24 is directly connected, GigabitEthernet0/0
L      172.16.1.1/32 is directly connected, GigabitEthernet0/0
R      172.16.2.0/24 [120/1] via 209.165.200.226, 00:00:12, Serial0/0/0
R      172.16.3.0/24 [120/2] via 209.165.200.226, 00:00:12, Serial0/0/0
R      172.16.4.0/28 [120/2] via 209.165.200.226, 00:00:12, Serial0/0/0
R      192.168.0.0/16 [120/2] via 209.165.200.226, 00:00:03, Serial0/0/0
        209.165.200.0/24 is variably subnetted, 5 subnets, 2 masks
C      209.165.200.224/30 is directly connected, Serial0/0/0
L      209.165.200.225/32 is directly connected, Serial0/0/0
R      209.165.200.228/30 [120/1] via 209.165.200.226, 00:00:12, Serial0/0/0
C      209.165.200.232/30 is directly connected, Serial0/0/1
L      209.165.200.233/32 is directly connected, Serial0/0/1
R1#
```

Dynamically Learned IPv4 Routes

Ultimate Route



Strathmore
UNIVERSITY

- An ultimate route is a routing table entry that contains either a next-hop IPv4 address or an exit interface.
- Directly connected, dynamically learned, and local routes are ultimate routes.

```
R1# show ip route | begin Gateway
Gateway of last resort is 209.165.200.234 to network 0.0.0.0

S*   0.0.0.0/0 [1/0] via 209.165.200.234, Serial0/0/1
      is directly connected, Serial0/0/1
      172.16.0.0/16 is variably subnetted, 5 subnets, 3 masks
C     172.16.1.0/24 is directly connected, GigabitEthernet0/0
L     172.16.1.1/32 is directly connected, GigabitEthernet0/0
R     172.16.2.0/24 [120/1] via 209.165.200.226, 00:00:12, Serial0/0/0
R     172.16.3.0/24 [120/2] via 209.165.200.226, 00:00:12, Serial0/0/0
R     172.16.4.0/28 [120/2] via 209.165.200.226, 00:00:12, Serial0/0/0
R     192.168.0.0/16 [120/2] via 209.165.200.226, 00:00:03, Serial0/0/0
      209.165.200.0/24 is variably subnetted, 5 subnets, 2 masks
C     209.165.200.224/30 is directly connected, Serial0/0/0
L     209.165.200.225/32 is directly connected, Serial0/0/0
R     209.165.200.228/30 [120/1] via 209.165.200.226, 00:00:12, Serial0/0/0
C     209.165.200.232/30 is directly connected, Serial0/0/1
L     209.165.200.233/32 is directly connected, Serial0/0/1
R1#
```

Dynamically Learned IPv4 Routes

Level 1 Route

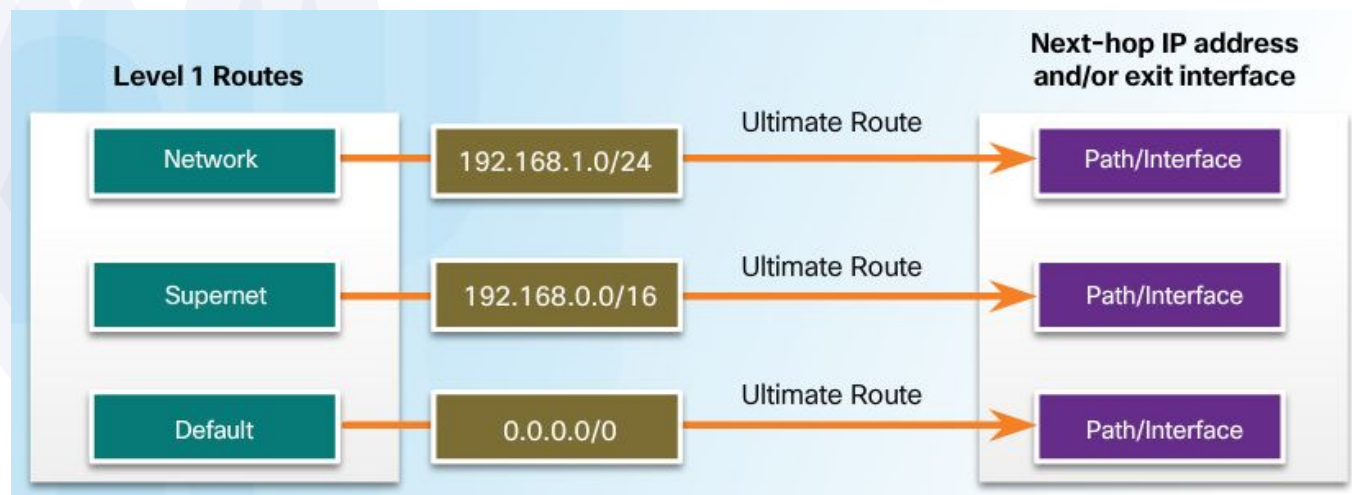


Strathmore
UNIVERSITY

- A level 1 route can be a:
 - **Network route** - a network route that has a subnet mask equal to that of the classful mask.
 - **Supernet route** - a network address with a mask less than the classful mask, for example, a summary address.
 - **Default route** - a static route with the address 0.0.0.0/0

```
R1# show ip route | begin Gateway
Gateway of last resort is 209.165.200.234 to network 0.0.0.0

S*   0.0.0.0/0 [1/0] via 209.165.200.234, Serial0/0/1
      is directly connected, Serial0/0/1
C     172.16.0.0/16 is variably subnetted, 5 subnets, 3 masks
C     172.16.1.0/24 is directly connected, GigabitEthernet0/0
L     172.16.1.1/32 is directly connected, GigabitEthernet0/0
R     172.16.2.0/24 [120/1] via 209.165.200.226, 00:00:12, Serial0/0/0
R     172.16.3.0/24 [120/2] via 209.165.200.226, 00:00:12, Serial0/0/0
R     172.16.4.0/28 [120/2] via 209.165.200.226, 00:00:12, Serial0/0/0
R     192.168.0.0/16 [120/2] via 209.165.200.226, 00:00:03, Serial0/0/0
C     209.165.200.0/24 is variably subnetted, 5 subnets, 2 masks
C     209.165.200.224/30 is directly connected, Serial0/0/0
L     209.165.200.225/32 is directly connected, Serial0/0/0
R     209.165.200.228/30 [120/1] via 209.165.200.226, 00:00:12, Serial0/0/0
C     209.165.200.232/30 is directly connected, Serial0/0/1
L     209.165.200.233/32 is directly connected, Serial0/0/1
R1#
```

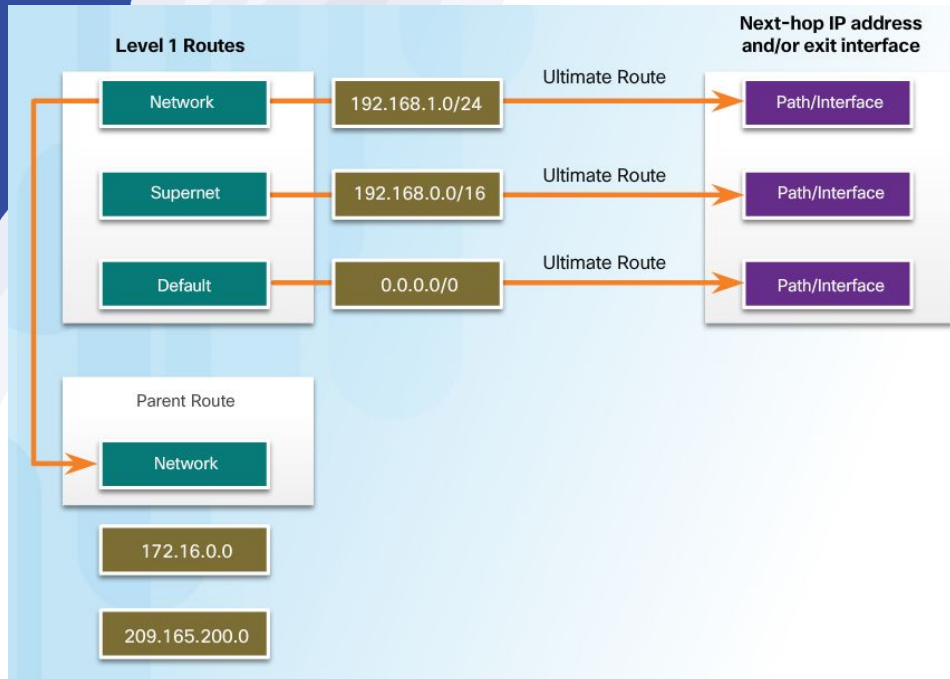


Dynamically Learned IPv4 Routes

Level 1 Parent Route



Strathmore
UNIVERSITY



- A parent route is a level 1 network route that is subnetted.
- In the routing table, it basically provides a heading for the specific subnets it contains.

```
R1# show ip route | begin Gateway
Gateway of last resort is 209.165.200.234 to network 0.0.0.0

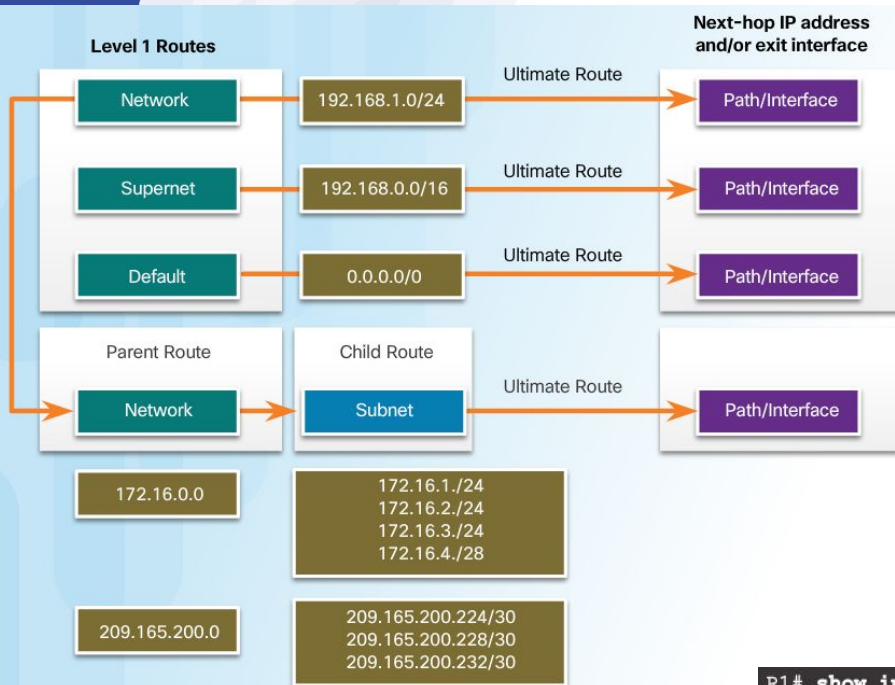
S*   0.0.0.0/0 [1/0] via 209.165.200.234, Serial0/0/1
      is directly connected, Serial0/0/1
C     172.16.0.0/16 is variably subnetted, 5 subnets, 3 masks
C     172.16.1.0/24 is directly connected, GigabitEthernet0/0
L     172.16.1.1/32 is directly connected, GigabitEthernet0/0
R     172.16.2.0/24 [120/1] via 209.165.200.226, 00:00:12, Serial0/0/0
R     172.16.3.0/24 [120/2] via 209.165.200.226, 00:00:12, Serial0/0/0
R     172.16.4.0/28 [120/2] via 209.165.200.226, 00:00:12, Serial0/0/0
R     192.168.0.0/16 [120/2] via 209.165.200.226, 00:00:03, Serial0/0/0
C     209.165.200.0/24 is variably subnetted, 5 subnets, 2 masks
C     209.165.200.224/30 is directly connected, Serial0/0/0
L     209.165.200.225/32 is directly connected, Serial0/0/0
R     209.165.200.228/30 [120/1] via 209.165.200.226, 00:00:12, Serial0/0/0
C     209.165.200.232/30 is directly connected, Serial0/0/1
L     209.165.200.233/32 is directly connected, Serial0/0/1
R1#
```

Dynamically Learned IPv4 Routes

Level 2 Child Route



Strathmore
UNIVERSITY



- A level 2 child route is a route that is a subnet of a classful network address.
- Level 1 parent routes contain level 2 child routes.
- Level 2 child routes are also ultimate routes.

```
R1# show ip route | begin Gateway
```

```
Gateway of last resort is 209.165.200.234 to network 0.0.0.0
```

```
S* 0.0.0.0/0 [1/0] via 209.165.200.234, Serial0/0/1
    is directly connected, Serial0/0/1
    172.16.0.0/16 is variably subnetted, 5 subnets, 3 masks
C   172.16.1.0/24 is directly connected, GigabitEthernet0/0
L   172.16.1.1/32 is directly connected, GigabitEthernet0/0
R   172.16.2.0/24 [120/1] via 209.165.200.226, 00:00:12, Serial0/0/0
R   172.16.3.0/24 [120/2] via 209.165.200.226, 00:00:12, Serial0/0/0
R   172.16.4.0/28 [120/2] via 209.165.200.226, 00:00:12, Serial0/0/0
R   192.168.0.0/16 [120/2] via 209.165.200.226, 00:00:03, Serial0/0/0
    209.165.200.0/24 is variably subnetted, 5 subnets, 2 masks
C   209.165.200.224/30 is directly connected, Serial0/0/0
L   209.165.200.225/32 is directly connected, Serial0/0/0
R   209.165.200.228/30 [120/1] via 209.165.200.226, 00:00:12, Serial0/0/0
C   209.165.200.232/30 is directly connected, Serial0/0/1
L   209.165.200.233/30 is directly connected, Serial0/0/1
```

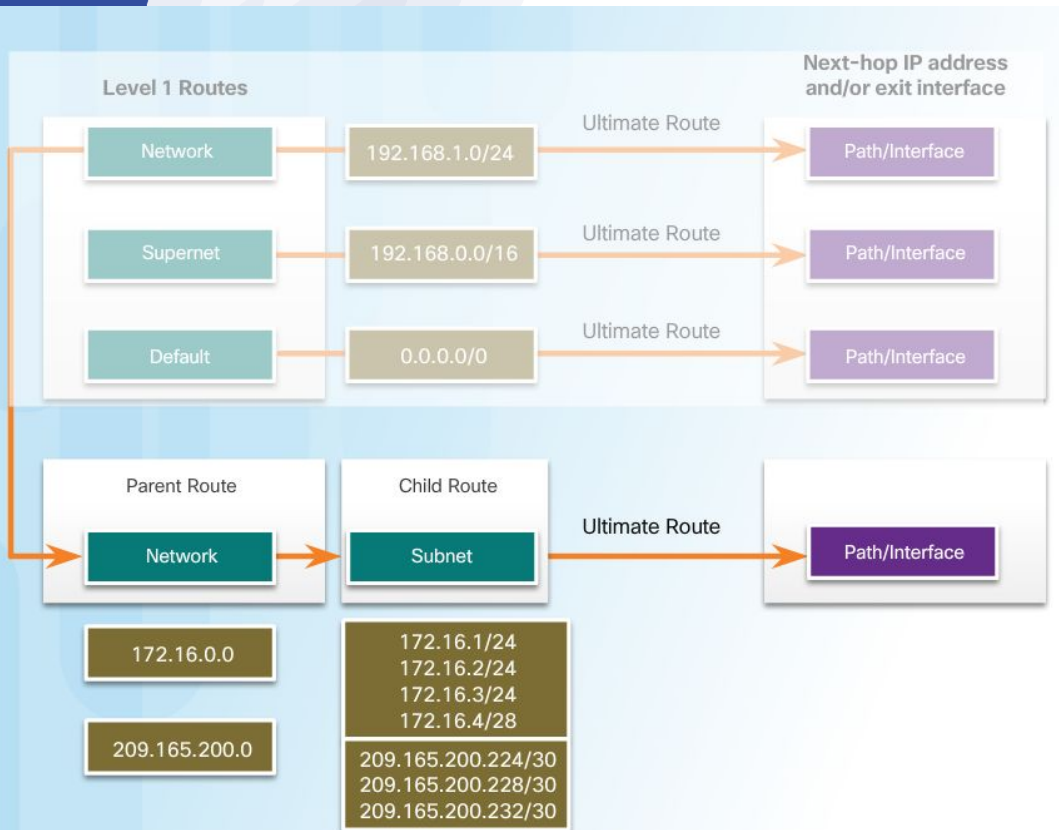
```
R1#
```

The IPv4 Route Lookup Process

Route Lookup Process



Strathmore
UNIVERSITY



- Router lookup process:
 - If the best match is a level 1 ultimate route, then this route is used to forward the packet.
 - If the best match is a level 1 parent route, the router then examines child routes (the subnet routes).
 - If there is a match with a level 2 child route, that is used to forward the packet.
 - If there is no match with level 2 child routes, the router searches level 1 supernet or default routes. If there is a match, that route is used.
 - If there is no match found in the routing table the packet is dropped.

The IPv4 Route Lookup Process

Best Route = Longest Match



Strathmore
UNIVERSITY

- The best match is the route in the routing table that has the most number of far left matching bits with the destination IPv4 address of the packet.
- The route with the greatest number of equivalent far left bits, or the longest match, is always the preferred route.

IP Packet Destination	172.16.0.10	10101100.00010000.00000000.00001010
Route 1	172.16.0.0/12	10101100.00010000.00000000.00000000
Route 2	172.16.0.0/18	10101100.00010000.00000000.00000000
Route 3	172.16.0.0/26	10101100.00010000.00000000.00000000

Longest Match to IP Packet Destination

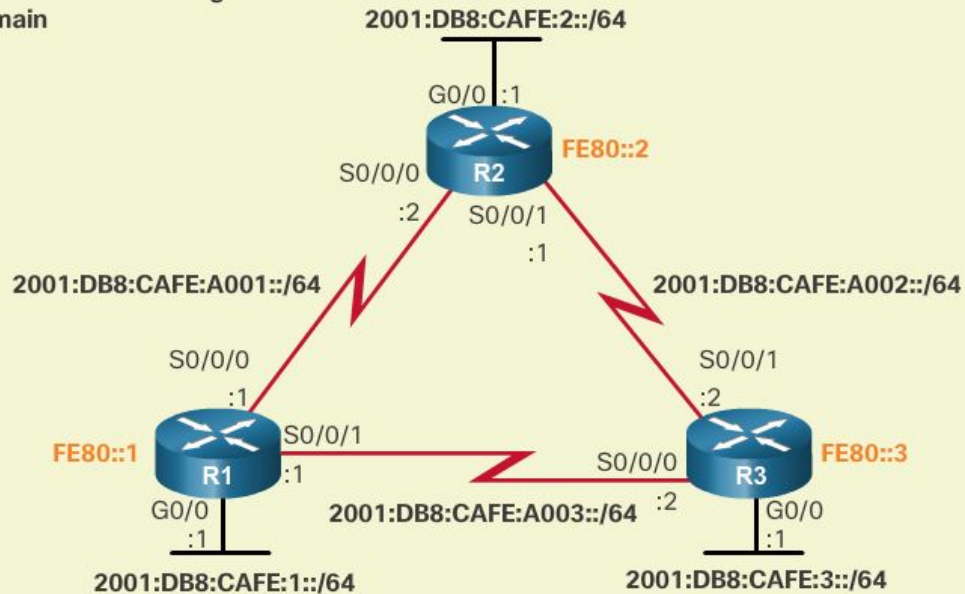
Analyze an IPv6 Routing Table

IPv6 Routing Table Entries



Strathmore
UNIVERSITY

EIGRP for IPv6 Routing
Domain



- An IPv6 routing table includes directly connected, static and dynamically learned routes.
- All IPv6 routes are level 1 ultimate routes.

The FE80 address represents the link-local address assigned to each router.

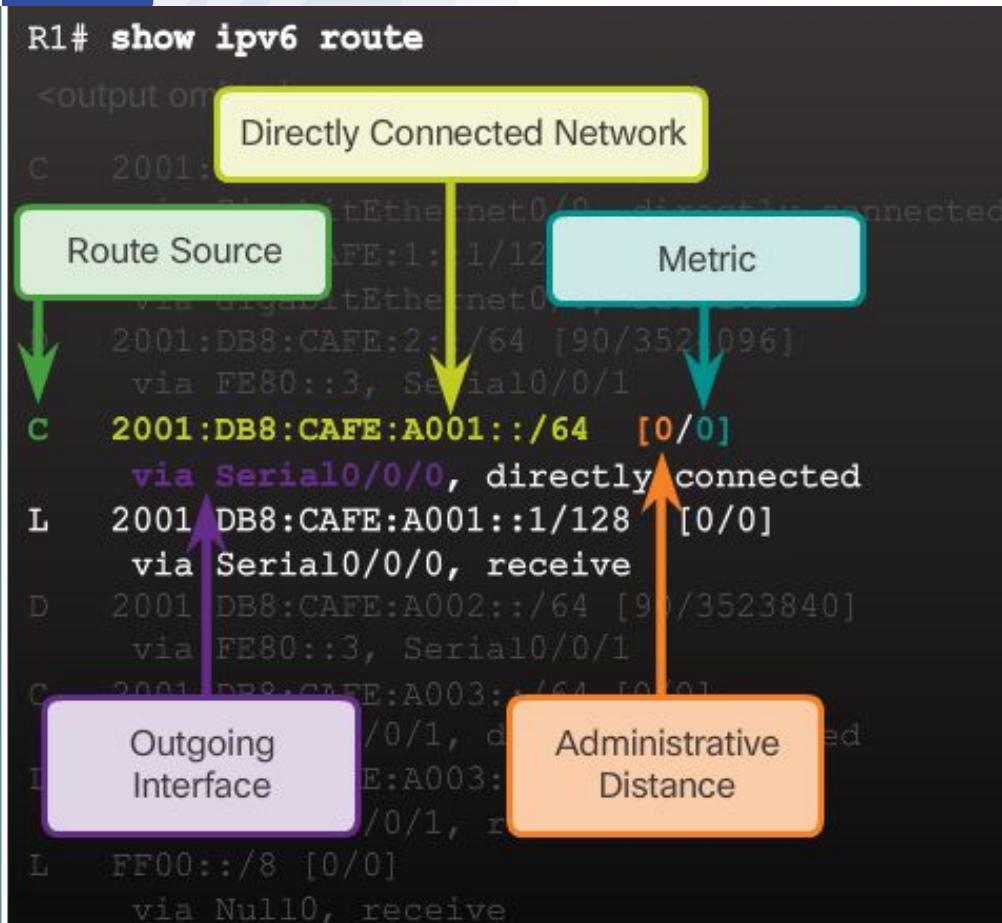
Analyze an IPv6 Routing Table

Directly Connected Entries



Strathmore
UNIVERSITY

- Use the **show ipv6 route** command to display the IPv6 routing table.
- The directly connected route entries include the following:
 - Route source – How the route was learned. Directly connected indicated with a C and L for local route.
 - Directly connected network address.
 - Administrative distance – Trustworthiness of the route (lower more trustworthy).
 - Metric – Value assigned to reach the network (lower is preferred route).
 - Outgoing interface – Exit interface used to forward packet.



Analyze an IPv6 Routing Table

Remote IPv6 Network Entries

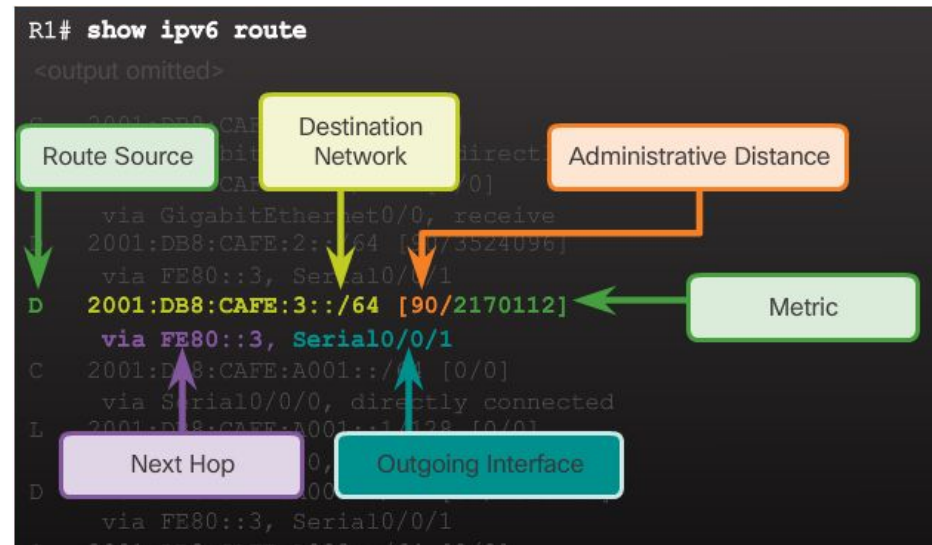


Strathmore
UNIVERSITY

- The remote IPv6 route entries also include the following:
 - Route source – How the route was learned. Common codes include O (OSPF), D (EIGRP), R (RIP), and S (Static route).
 - Next hop - Identifies the IPv6 address of the next router to forward the packet to.

The IPv6 router lookup process:

- Examines level 1 network routes for the best match.
- Longest match is the best match.



Routing Table Terms



Strathmore
UNIVERSITY

- The routing table is a hierarchical structure that is used to speed up the lookup process when locating routes and forwarding packets.
- The hierarchy includes:
 1. Ultimate Routes
 2. Level 1 routes
 3. Level 2 child routes

Routing Table Terms



1. Ultimate Routes

- a routing table entry that contains either the **IP address** of the **next-hop** or an **exit interface** or **both**
- Directly connected, dynamically learned, and local routes are ultimate routes

2. A Level 1 Route can be a

- Network route** - a network route that has a subnet mask equal to that of the classful mask
- Supernet route** - a network address with a mask less than the classful mask, for example, a summary address
- Default route** - a static route with the address 0.0.0.0/0
- Parent Route** - An entry in the routing table that has no **IP address** of the **next-hop** or an **exit interface**
 - A parent route provides a heading for the specific subnets it contains
 - The subnets are called **child routes**

3. Level 2 Child Routes

- A level 2 child route is a routing table entry that is a subnet of a classful network address
- Level 2 child routes are also **ultimate routes**

Level 1 Parent Route

Level 2 Child Routes,
Ultimate Routes

	172.16.0.0/24 is subnetted, 3 subnets
\$	172.16.1.0 [1/0] via 172.16.2.2
C	172.16.2.0 is directly connected, Serial0/0/0
C	172.16.3.0 is directly connected, FastEthernet0/0
\$	192.168.1.0/24 [1/0] via 172.16.2.2
\$	192.168.2.0/24 is directly connected, Serial0/0/0
S*	0.0.0.0/0 is directly connected, FastEthernet0/0

Level 1 routes,
Ultimate routes

Router Lookup Process

- When a packet is sent to a router for routing the router examines its routing table by comparing the destination address in the packet with the routing table entries in a sequential manner
- The subnet mask in the routing table is used to determine the minimum number of bits that must match between the destination address and the entry in the routing table for the data to be forwarded on the exit interface or to the next hop address that is indicated
- To find the best match the router tries to match the bits beyond the minimum required

NB: The best route is the one

- i. that has the longest number of leftmost matching bits and
- ii. is an ultimate route

Router Lookup Process



- **Example:**

Matches for Packet Destined to 172.16.0.10

IP Packet Destination	172.16.0.10	10101100.00010000.00000000.00001010
Route 1	172.16.0.0/12	10101100.00010000.00000000.00000000
Route 2	172.16.0.0/18	10101100.00010000.00000000.00000000
Route 3	172.16.0.0/26	10101100.00010000.00000000.00000000

Longest Match to IP Packet Destination

- How the hierarchical routing table is searched:
 - If the best match is a level 1 ultimate route, then this route is used to forward the packet
 - If the best match is a level 1 parent route, the router then examines child routes (the subnet routes)
 - If there is a match with a level 2 child route, that is used to forward the packet
 - If there is no match with level 2 child routes, the router searches level 1 supernet or default routes. If there is a match, that route is used
 - If there is no match found in the routing table the packet is dropped

Practice Exercise



- Identify the following routing table entries: level 1 routes, level 1 parent routes, level 2 child routes, ultimate routes
- Explain THREE things that can be concluded from the routing table entries shown in the figure below.
- Sketch a simple diagram to illustrate the topology shown in the table below.
- **Note:** *The topology will be incomplete since you have partial information*

```
Rawalpindi#show ip route
```

```
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP  
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area  
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2  
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP  
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area  
* - candidate default, U - per-user static route, o - ODR  
P - periodic downloaded static route
```

```
Gateway of last resort is not set
```

```
D    192.168.0.0/24 [90/2172416] via 192.168.10.1, 00:00:20, Serial0/3/0  
D    192.168.1.0/24 [90/2172416] via 192.168.10.1, 00:00:20, Serial0/3/0  
C    192.168.3.0/24 is directly connected, FastEthernet0/0  
C    192.168.4.0/24 is directly connected, FastEthernet0/1  
      192.168.10.0/30 is subnetted, 1 subnets  
C      192.168.10.0 is directly connected, Serial0/3/0
```